Arunabha Majumder

Mechanical Engineer | Robot Integration, Testing and Calibration

+45 71 62 24 00 | Aalborg, Denmark | somrkmv1997@gmail.com | arunabha22.github.io/arunabha-majumder

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Aperion Robotics, iKnowHow Group

Application: Mechanical Engineer, Production Division

Dear Hiring Team,

I am applying for the Mechanical Engineer position in your Production Division. I am a mechatronics engineer finishing my PhD at Aalborg University, with an MTech in Mechatronics and a BTech in Mechanical Engineering. The role caught my attention because it sits exactly where I have spent most of my time: between the mechanical parts, the electronics and the software, making sure a robot actually works when it leaves the bench.

In my PhD I built a shoulder exoskeleton and integrated the motor, encoder and force sensors with the mechanical structure myself. I designed the parts in SolidWorks, 3D printed them and worked with two colleagues on the controller, where my part was the equations, such as a Fourier fit, and deciding how the GUI should behave. A lot of the work was debugging: loose connections, sensor offsets and code that behaved differently on the real hardware. The design cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026.

I also do a lot of testing and calibration. On a 5-bar parallel robot I estimated the force at the tip from variable-stiffness actuators and checked it against a reference force sensor, getting within 10% RMS error. I built and maintained test rigs, wrote test procedures and collected data from 12 participants. I use Python and MATLAB for data and testing, and I keep clear notes so that someone else can repeat a setup.

I want to be honest about the gaps. I have not yet worked with ROS or done commissioning at a customer site. I learn new tools quickly, and I would put ROS and Linux at the top of my list in the first months. Being the link between production, engineering and software is something I already do in the lab, and I have taught and supervised students since 2023, so explaining technical problems clearly to different people comes naturally. I am also happy to travel to customer sites.

Thank you for considering my application. I would be glad to talk about how I can help get your robots integrated, tested and running for your customers.

Kind regards,
Arunabha Majumder